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The impact of childhood trauma on borderline personality organization in a community sample of Greek emerging adults

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ABSTRACT

There is limited research on the role of childhood trauma in personality pathology according to Kernberg's psychodynamic model of internalized object relations. Because childhood trauma reflects the disruptions of these relations, it is expected to predict borderline personality organization, especially at the threshold of adulthood. Therefore, the main aim of this retrospective study was to examine the impact of childhood trauma on borderline personality organization in a community sample of emerging adults. Participants were 543 Greek individuals aged 18–29 ($M = 21.45$; 58.6 % females; 85.1 % university students). They completed the Greek versions of the Traumatic Antecedents Questionnaire (TAQ) and the Inventory of Personality Organization (IPO), which were tested for their factorial structure, reliability, and measurement invariance across gender, as few empirical data exist on the psychometric properties of these measures. Confirmatory factor analyses showed that the TAQ consisted of four factors, namely positive experiences, abuse, traumatic life events, and family chaos. The five-factor theoretical structure of the IPO, namely primitive defenses, identity diffusion, reality testing, aggression, and moral values, was confirmed. Low to moderate links between childhood trauma and borderline personality organization were found, with stronger links emerging for abuse and family chaos. Structural equation modeling showed that the various forms of childhood trauma across the age periods studied (i.e., 0–6, 7–12, 13–18) significantly and differentially predicted the dimensions of borderline personality organization. The finding that stronger links emerged when trauma occurred in older ages may be attributed to the retrospective method of the study. Gender differences were also found; for example, personality pathology was more likely in men when abuse and traumatic life events occurred in younger ages and abuse was a more important risk factor for personality pathology in women. This study highlights the impact of childhood adversity on personality pathology in emerging adulthood, provides empirical support for Kernberg's psychodynamic model, and has useful implications for trauma-informed early screening, prevention, and intervention regarding personality pathology in young people. Limitations of this study and suggestions for future research are outlined.

1. Introduction

A large body of research, including both retrospective and prospective studies in clinical and nonclinical populations, has supported the significant impact of childhood adversity on personality disorders. Yet, there are few studies to date examining the role of childhood trauma in borderline personality organization (BPO) according to Otto F. Kernberg's (1976) psychodynamic model of internalized object relations, which are defined as units of self-image, object-image and the affectively-charged interaction between them. The internalization of object relations begins early in life as the infant interacts with significant others, continues with further modification and organization in childhood and adolescence, and forms the basis of personality development.

Although personality development and pathology are determined by

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a dynamic interaction of genetic/neurobiological, psychodynamic, and psychosocial factors according to this model (Kernberg, 2015), the internalized object relations are of crucial importance among these factors. Within this evidence-based, constantly evolving model of personality pathology, BPO is conceptualized as a pathological psychic structure – the other two being the neurotic and the psychotic levels of organization – which includes most personality disorders and has various interrelated features (Kernberg, 1975, 1984; see also Caligor et al., 2023; Clarkin et al., 2007): identity diffusion (i.e., representations of self and others are inconsistent, unintegrated, unstable, unrealistic or even distorted, with their emotional charge being shallow, intense or extreme); primitive defense mechanisms in coping with external stress and inner conflict (e.g., splitting, projection, projective identification, primitive idealization, omnipotent control); low quality of internalized object relations (i.e., limited depth, mutuality, and concern, tendency to explore others); difficulties in cognitive and emotion regulation (i.e., instability, mood swings); difficulties in empathy, mentalization, and intimacy; impaired capacity to identify with and pursue long-term goals and personal interests; low internalization of a system of moral values (i.e., deficits in superego formation, limited moral functioning, lack of guilt, antisocial behavior); aggression (i.e., verbal and/or physical aggression towards self and others); and impaired reality testing (i.e., ranging from mild difficulties in reading social cues and perceiving social conventions and others' internal states to psychotic distortions, especially under stress).

Due to its psychoanalytic origin, Kernberg's model of internalized object relations emphasizes the role of early traumatic or adverse experiences in the development of the BPO. Trauma is conceptualized as situations or events that overwhelm individuals by causing excessive excitation and exceeding their capacity to tolerate, control, and work through it, thus having persistent negative consequences on their mental health (Laplanche & Pontalis, 1967). Childhood trauma occurs before the age of 18 and takes on various forms, for example, abuse (physical, sexual, emotional, verbal) and neglect (physical, emotional), maladaptive parenting, negative life events (e.g., accidents, disasters), etc. The negative consequences of childhood trauma on the psychosomatic health and well-being of the developing individual are well documented (Devi et al., 2019; Humphreys et al., 2020; see also van der Kolk, 2014). Moreover, epidemiological studies have indicated the high prevalence of several forms of childhood maltreatment (e.g., Curran et al., 2018; Saunders & Adams, 2014).

Therefore, this study examines the associations between various forms of childhood trauma and BPO by focusing on emerging adulthood (ages 18–29), because this period is the age of onset of clinically defined personality disorders (Chanen & Thompson, 2019), while it is also characterized by a "borderline" feeling per se of being neither adolescent nor adult (Arnett, 2015). In addition, Kernberg's model, as the theoretical framework of this study, is expected to provide useful clinical implications for at least two reasons: (a) it emphasizes the crucial role of the disruption of internalized object relations – which is typically caused by early trauma – in the emergence of personality pathology; and (b) it has greatly inspired the recent shift in the conceptualization and treatment of mental disorders towards a dimensional model of personality pathology, as evidenced in the Psychodynamic Diagnostic Manual (2nd ed.; Lingiardi & McWilliams, 2017) and in the Alternative DSM-5 Model for Personality Disorders (included in the Diagnostic and Statistical Manual - 5th ed.; American Psychiatric Association, 2013).

1.1. Childhood trauma and personality disorders

Existing retrospective and prospective research in clinical and nonclinical samples shows that various types of childhood trauma, such as physical abuse, sexual abuse, emotional abuse, emotional neglect, physical neglect, and maladaptive parenting (e.g., rejection, overcontrol, invalidation) are associated with later development of various personality disorders, such as borderline, narcissistic, schizotypal, and antisocial personality disorder (for reviews see Carr et al., 2013; Perry & Lee, 2020; Schorr et al., 2020; Xiao et al., 2023). For example, childhood adverse experiences overall, as assessed either with objective measures or retrospectively, were associated with four-times greater risk for personality disorders in early adulthood (Johnson et al., 1999). Childhood abuse and neglect, particularly, were the forms of childhood adversity mostly related to several personality disorders (Afifi et al., 2011). Sexual abuse was the type of abuse associated with the largest range of personality disorders in a nonclinical university student sample (Gibb et al., 2001). Furthermore, in other studies (Cohen et al., 2014; Lobbestael et al., 2010), different types of childhood maltreatment predicted different personality disorders. For example, it has been found that physical abuse was associated with antisocial personality disorder and sexual abuse with paranoid, schizoid, borderline, and avoidant personality disorders.

Likewise, there are many studies examining the links of childhood trauma with borderline personality disorder, specifically, yielding mixed results regarding the type of abuse that is most strongly associated with this disorder, perhaps due to measurement and methodological differences. Relevant reviews (Bozzatello et al., 2021; de Aquino Ferreira et al., 2018; Hutsebaut & Aleva, 2021; Martins et al., 2011; Stepp et al., 2016; Yuan, Lee, Eack, et al., 2023) and meta-analyses (Porter et al., 2020; Winsper et al., 2016) have shown that sexual abuse, either from the caregiver or other person, significantly predicts borderline personality disorder. It has also been found that this association is stronger for sexual abuse than other forms of abuse (Hailes et al., 2019). However, the percentages of people diagnosed with borderline personality disorder who have a history of sexual abuse vary considerably in several studies (de Aquino Ferreira et al., 2018). An earlier review indicated that borderline personality disorder was predicted by sexual and physical abuse, mother hostility and verbal abuse, as well as neglect (Winsper et al., 2016). A more recent review revealed that emotional abuse and sexual abuse were the first and second most significant predictors, followed by physical abuse, emotional neglect, and physical neglect (Yuan, Lee, Eack, et al., 2023). Along these lines, in another recent meta-analysis (Porter et al., 2020) emotional abuse and neglect were found to be the main predictive factors of this disorder, and this prediction was stronger for clinical than nonclinical samples.

A closer look at the literature reveals robust links of borderline personality disorder with sexual abuse in childhood (e.g., Fossati et al., 2016; Johnson et al., 1999; Paris et al., 1994; Zanarini et al., 1997), but also with physical abuse (e.g., Battle et al., 2004; Carlson et al., 2009; Sansone et al., 2005; Widom et al., 2009) as well as with emotional and verbal abuse (e.g., Bornovalova et al., 2013; Kim et al., 2018; Rosenstein et al., 2018; Waxman et al., 2014; Yuan, Lee, Newhill, et al., 2023), neglect (e.g., Johnson et al., 2000; Lyons-Ruth et al., 2013; Zanarini et al., 1997), exposure to domestic violence (Herman et al., 1989), and the combination of abuse and neglect (Brodbeck et al., 2018). In addition, dysfunctional parenting practices have been associated with borderline personality disorder (e.g., Cohen et al., 2005, 2008; Lyons-Ruth et al., 2013; Musser et al., 2018; Robertson et al., 2013). Early separation from parents (Bandelow et al., 2005; Crawford et al., 2009) and parental problems such as conflicts (e.g., Herman & van der Kolk, 1987; Porter et al., 2020; Weaver & Clum, 1993), psychopathology and hospitalization (e.g., Barnow et al., 2013; Conway et al., 2015; Reinelt et al., 2014), father drug use (Stepp et al., 2013), as well as childhood illnesses (Bandelow et al., 2005) were also predictors of borderline personality disorder. Importantly, empirical evidence supports the view that separation from parents and neglect take place within a pre-existing familial context of abuse (Zanarini et al., 1989).

On the contrary, there is recently growing evidence that positive or benevolent childhood experiences have significant links with positive outcomes for mental and physical health as well as for psychosocial functioning in adulthood (for a review see Han et al., 2023). Regarding personality pathology, in a retrospective study (Saleptsi et al., 2004), it was found that feelings of safety during childhood and adolescence were

lower in patients with personality disorders than in patients with other disorders or in controls. Prognosis of some personality disorders (i.e., schizotypal and avoidant) was better for psychiatric patients who, during their childhood, had experienced achievements, positive interpersonal relationships, and caretaker competencies (Skodol et al., 2007). In another study (Gunay-Oge et al., 2020), benevolent childhood experiences, that is, perceived safety (e.g., from caregivers), positive and predictable quality of life (e.g., enjoyment of school), and interpersonal support (e.g., a caring teacher), were less likely among young adults from nonclinical population who exhibited 11 out of 14 personality disorder symptoms (including borderline personality disorder). In sum, there is limited research on the role of positive childhood experiences in the emergence of personality pathology.

1.2. Childhood trauma and Kernberg's borderline personality organization

Only few empirical studies have examined childhood traumatic experiences as predictors of personality disorders within the dimensional model (i.e., Alternative Model of Personality Disorders, AMPD) introduced in DSM-5 (American Psychological Association, 2013). Based on the Inventory of Personality Organization (IPO), which was constructed within Kernberg's framework to assess borderline personality organization (Kernberg & Clarkin, 1995), these studies showed that emotional abuse and neglect are associated with personality dysfunction and with maladaptive personality traits, mainly detachment and psychoticism, more than physical and sexual abuse (for a review see Back et al., 2021).

Particularly, in a university student sample in Japan, neglect, emotional abuse and, to a lesser extent, sexual abuse and punishment had low to moderate links with BPO, as assessed by the full version of the IPO, that is, with the subscales named primitive defenses, identity diffusion, reality testing, aggression, and moral values (Igarashi et al., 2010). Moreover, BPO in university students in Japan, as assessed again by the full version of the IPO, was related to their tendency to perceive and create negative life events (Uji et al., 2013). With the three primary subscales of the IPO, that is, primitive defenses, identity diffusion, and reality testing, there were low to relatively moderate associations of childhood adversity and various types of adult trauma with BPO in adult patients in the USA (Espinosa & Rudenstine, 2018). Finally, in community samples of young adults and adults in Germany (Fuchshuber et al., 2018, 2019), low to moderate links were identified between childhood trauma, mainly abuse and neglect, and BPO, as assessed by a brief version of the IPO.

Consequently, there are very few existing data on the predictive role of childhood trauma in the emergence of BPO according to Kernberg's psychodynamic framework and with the use of the IPO and, to our knowledge, there is no study examining positive childhood experiences in relation to BPO. Moreover, the theoretical structure of the IPO appears to warrant further empirical confirmation in populations of different ages and different cultural backgrounds. It is especially important to examine young people's personality pathology in various cultural contexts because existing research (e.g., Žukauskienė, 2016; for Greek emerging adults see Galanaki & Sideridis, 2019) has shown that emerging adulthood is a developmental period that is largely determined by cultural factors and the pathways to adulthood are highly diverse depending on the cultural context.

2. Aims and hypotheses

The principal aim of this retrospective study is to examine whether and to what extent childhood trauma can predict BPO during emerging adulthood (ages 18–29). This developmental period is characterized by identity exploration, experimentation and open possibilities, self-focus, instability, and an in-between or "borderline" feeling of being neither adolescent nor adult, since young people feel on the threshold of adulthood, but not there yet (Arnett, 2015; Arnett et al., 2014). The in-between feeling has been found to be highly prevalent in Greek emerging adults (Galanaki & Sideridis, 2019). This period is also the age of onset of personality disorders as clinical entities, although often dysfunctional and maladaptive personality traits are already evident during childhood (Chanen & Thompson, 2019). Personality pathology, even if it does not meet the criteria for a diagnosis of personality disorder according to the categorical model and rather manifests itself with subclinical/subthreshold features, impedes adult role taking (Chanen et al., 2020). Adult functioning presupposes the formation of a coherent, stable, and realistic identity as well as the capacity for relationships high in intimacy and cooperativeness (Erikson, 1963). However, these developmental milestones are jeopardized by personality pathology. Indeed, previous research has shown that typical features of borderline personality disorder and BPO, such as anger dyscontrol, emotional lability, impulsive behaviors, difficulties in relationships, self-harm, and other self-destructive behaviors, are common among university students (e.g., Gratz et al., 2002; Trull, 1995; Trull et al., 1997). Moreover, emerging adults are expected to be able to provide reliable and valid retrospective reports of their personal history. This happens not only because they possess the necessary cognitive and metacognitive abilities, such as reflection, perception of complex associations, searching for and finding meaning in life, and ability to construct autobiographical narratives (King & Kitchener, 2016), but also because, by being temporally close to their childhood and adolescence, they can recall experiences from these age periods.

The fact that Kernberg's model of BPO places emphasis on the importance of object relations for personality development and pathology indicates the need to examine the associations of BPO with childhood experiences, as these experiences determine the quality of internalized relations with significant others (Kernberg, 1976, 2015). We therefore hypothesize that positive childhood experiences will predict lower levels of BPO, whereas traumatic experiences will predict higher levels. Stronger links with BPO are expected to be found for traumatic experiences that are related to direct disruptions of object relations, that is, for all types of abuse and neglect as well as family dysfunction, in accordance with existing data (e.g., Bozzatello et al., 2021; de Aquino Ferreira et al., 2018; Hutsebaut & Aleva, 2021; Porter et al., 2020; Yuan, Lee, Eack, et al., 2023). This hypothesis is also based on the fact that the disruption of attachment leads to disruptions of internalized object relations, which set the stage for BPO (Clarkin et al., 2007), and that primitive defense mechanisms (e.g., splitting, projection) are activated as the individual attempts to avoid the painful awareness of the traumatic relationships with significant others (Caligor & Clarkin, 2010). These experiences are regarded as *interpersonal trauma* and, if multiple and chronic, they result in the developmental trauma disorder (Spinazzola et al., 2021; van der Kolk, 2005). In contrast, less strong are expected to be the links with non-interpersonal trauma, that is, traumatic life events, which are only indirectly related to disruptions in relationships with significant others, thus they are considered as not necessarily impacting the internalization of object relations. In addition, because Kernberg's (1976) psychodynamic model emphasizes the decisive role of early experiences for personality development and pathology – given that object relations are internalized from the beginning of life –, we anticipate finding stronger links between childhood trauma and BPO when trauma has occurred in younger ages.

Although this research includes a community sample, we expect that positive links between childhood trauma and BPO will be identified. However, consistent with data from other studies that have examined BPO in community samples of young people, especially university students (Fuchshuber et al., 2018, 2019; Igarashi et al., 2010), we expect that these links will be low to moderate.

3. Method

3.1. Participants and procedure

Participants were 543 individuals aged 18–29 (M = 21.45, SD = 2.51) representing the full age range of emerging adulthood. Of them 85.1 % were university students, 58.6 % were females, and 65 % had no job. Regarding their residence, 22.3 % lived alone, 57.3 % lived with their parents, 10.1 % lived with other relatives, 8.3 % lived with friend or romantic partner, and 2 % had other living arrangement. As to their relationship status, 45.1 % had no romantic relationship, 12.5 % had casual romantic relationships, 40 % had a stable romantic relationship, 2 % were in marriage or civil partnership, and 0.4 % were divorced. Moreover, currently and during the previous one year, 29.1 % had sought help from mental health professionals, 25.4 % had received counseling or psychotherapy, 6.1 % had experience of medication for mental health problems, and 16 % had a chronic somatic illness (e.g., asthma).

Recruitment of participants was done through an invitation for voluntary participation and completion of the instruments in electronic form (via Google Forms). After university teachers' permission, the link was uploaded on e-class platforms of several university departments located in Athens. The link was also posted on social media groups of university students from institutions all over Greece. Through snowball sampling, participants were then asked to forward the link to other individuals aged 18–29.

Participants were informed about the aim of the research and their rights (e.g., to withdraw from the study at any time), were ensured for the anonymity and confidentiality of their personal data, given the sensitive nature of the topic, and then provided informed consent. This study is part of a research project on personality disorders carried out by the Personality Disorders Specific Sector, 1st Psychiatric Clinic, Eginition Hospital, School of Medicine, National and Kapodistrian University of Athens. The project has been approved by the Research Ethics Committee of Eginitio Hospital (approval number: 404/5.7.2021).

3.2. Measures

Traumatic Antecedents Questionnaire (TAQ; Herman & van der Kolk, 1987; Spinazzola, 2019. In this study, the Greek version of the TAQ was used (Karamanolaki et al., 2016; Saleptsi et al., 2004). It is a self-report 41-item instrument which assesses individuals' recall of past and present positive and potentially traumatic experiences. Subscales include competence, safety, neglect, separation, emotional abuse, physical abuse, sexual abuse, witnessing trauma, other trauma, and alcohol and drug use. The last item assesses distress during the completion of the questionnaire and provides information for post-traumatic stress. Individuals were asked to recall these experiences for ages 0–6, 7–12, 13–18, and adulthood, and rate their frequency or intensity on a 4-point scale ranging from 0 (*never* or *not at all*) to 3 (*often* or *very much*). The *don't know* answers were coded as missing values. Given that this study focuses on childhood trauma, answers for only the three first age periods were analyzed. Very few studies have examined the factorial structure of the measure (Park et al., 2020; Saleptsi et al., 2004). In the study by Saleptsi et al. (2004), an exploratory factor analysis of the Greek version yielded five factors: traumatic experiences, positive experiences, vulnerability to alcohol abuse, disruptions of attachment, and family chaos. Adequate internal consistency reliability was found in another study with the Greek version (Cronbach's alpha = 0.80; Karamanolaki et al., 2016, where the factors of the Saleptsi et al.'s study were used). A detailed examination of the factorial structure, reliability, and measurement invariance of the Greek TAQ is presented in the Results.

Inventory of Personality Organization (IPO; Kernberg & Clarkin, 1995; Berghuis et al., 2009; Clarkin et al., 2001; Lenzenweger et al., 2001). It is a self-report 83-item inventory which screens for BPO according to Kernberg's object relations model. Subscales include primitive defenses, identity diffusion, reality testing, aggression, and moral values. Individuals were asked to indicate how each statement applies to them on a 5-point scale ranging from 1 (*never true*) to 5 (*always true*).

Studies using the full version of the IPO have indicated satisfactory internal consistency reliability for the total scale and its dimensions, but yielded mixed results about its factorial structure. Specifically, they have indicated three or four factors that are either similar (yet not identical) to the theoretical factors or mixed (Barreto et al., 2017; Cosentino et al., 2021; Berghuis et al., 2009; Ellison & Levy, 2012; Hörz-Sagstetter et al., 2021; Lenzenweger et al., 2001; Normandin et al., 2002; Preti et al., 2015; Smits et al., 2009· Verreault et al., 2013; Wang et al., 2022). Therefore, it was necessary to examine whether the theoretical structure of the measure is confirmed in the Greek sample. A detailed examination of the factorial structure, reliability, and measurement invariance of the Greek IPO are presented in the Results.

For the development of the Greek version of the IPO, initially the items were translated into Greek independently by the first and the fourth co-author. Then, consensus was attained with the participation of the fifth co-author. Finally, this version was back translated into English by a bilingual psychologist and minor linguistic adjustments were made to the final Greek version.

Sociodemographic information. Participants were asked to provide information about gender, age, education, profession, country of origin, family status, living arrangement, and relationship status. They also indicated whether in the last year prior to data collection had sought help from mental health professionals, had received counseling or psychotherapy, and had taken medication for mental health problems as well as whether they had a chronic somatic illness.

3.3. Data analysis

First, confirmatory factor analyses (CFAs) were conducted to test the factorial structure of the measures (Brown, 2015; Hu & Bentler, 1999; Kline, 2015). McDonald's omega (ω; Raykov, 1997) was used for the internal consistency reliability of the measures and Cronbach's alpha (α) for factors with only two items. Measurement invariance (i.e., configural, metric, and scalar) across gender of both measures was also examined. If measurement invariance tests fail, partial measurement invariance with the alignment method (Asparouhov & Muthén, 2014) is used. Invariant parameters across gender (slopes and intercepts) were evaluated using z-tests. For the main research question (i.e., the predictive role of childhood trauma in BPO), structural equation modeling (SEM), using maximum likelihood method with robust standard errors (MLR) for possible deviations from normality, was conducted. Estimates are standardized coefficients in the standard deviation metric. Based on Cohen's (1992) suggestions, coefficients 0.2, 0.5, and 0.8 indicate small, moderate, and large effect sizes, respectively. Missing values were treated in a pairwise fashion. Thus, all psychometric models had the full sample size with no exclusion for single participants unless they were missing across all items. Beyond psychometrics, predictive models utilized mean scores which were valid for all participants as they are estimated despite having missing values. Analyses were conducted with SPSS 28.01 and Mplus 8.10.

4. Results

4.1. Factorial structure of the measures and measurement invariance across gender

For the TAQ, to achieve a more parsimonious factor structure that could be used in further analyses, we initially conducted a series of exploratory factor analyses (principal component analyses with varimax rotation), which showed that a fixed four-factor structure was the most appropriate solution based on eigenvalues and percentages of explained

variance. Then, CFA indicated a good fit of the four-factor structure to the data: $\chi^2(29) = 43.482$, CFI = 0.954, TLI = 0.929, RMSEA = 0.039 (see also Supplementary File, Fig. S1). The factors were named as follows: positive childhood experiences (competence, safety; α = 0.63), abuse (emotional abuse, physical abuse, witnessing trauma; ω = 0.80), traumatic life events (separation, other trauma; α = 0.56), and family chaos (sexual abuse, neglect, alcohol and drug use; ω = 0.59). This factor structure is similar, though not identical, to the five-factor structure found by Saleptsi et al. (2004) and used by Karamanolaki et al. (2016) in Greece.

For the IPO, with the aim of confirming a factorial structure as close to the theoretical one as possible, we used item parcels to (a) reduce measurement error, (b) increase the likelihood that distributions will approximate normality, (c) reduce the number of estimated parameters, thus increase parsimony, and (d) aid model convergence (Little et al., 2002). The CFA indicated a good fit of the five-factor structure to the data: $\chi^2(142) = 380.120$, CFI = 0.905, TLI = 0.886, RMSEA = 0.056 (see also Supplementary File, Fig. S2). The five factors were identical to the theoretical ones: primitive defenses (ω = 0.79), identity diffusion (ω = 0.88), reality testing (ω = 0.81), aggression (ω = 0.81), and moral values (ω = 0.73).

The TAQ exhibited exact measurement invariance across gender, whereas for the IPO the alignment method (Asparouhov & Muthén, 2014) was applied to achieve invariance, so that latent mean comparisons would be possible (for more information see Supplementary File, Table S1). Thus, the two measures are equivalent across gender.

4.2. Descriptive statistics

In this community sample, the levels of traumatic experiences were low, as expected, whereas the levels of positive experiences were higher than those of traumatic experiences (see Table 1). Similarly, the levels of BPO were relatively low, and only the levels of identity diffusion were found to be higher than the levels of all other dimensions of BPO (> 2 in a 5-point scale).

4.3. Correlations between variables

First, there was a moderate positive correlation between the total scores of the TAQ and the IPO (Pearson $r = 0.40$, $p < .001$). Next, as shown in Table 2, there were statistically significant low to moderate positive correlations between the dimensions of the TAQ and the dimensions of the IPO. The larger correlations (≥ .|30|) were found between positive childhood experiences and identity diffusion, reality testing, and aggression (negative ones), between abuse and all dimensions of the IPO (positive ones), between traumatic life events and reality testing and aggression (positive ones), and between family chaos and aggression (positive ones). Finally, the question which assesses distress during the completion of the TAQ, as an indication of post-traumatic stress, had a low to moderate positive correlation with the total TAQ (Pearson $r = 0.31$, $p < .001$); the highest correlation was with abuse (Pearson $r = 0.32$, $p < .001$). All correlations of distress with total BPO and its five factors, although positive and statistically significant, were low (< 0.30).

Table 1
Descriptive statistics for all variables (N = 543).

Variables	*M*	*SD*	*Min*	*Max*
Traumatic Antecedents Questionnaire[a]	0.48	0.18	0.14	1.31
Positive experiences – Total	2.15	0.49	0.06	3.00
Positive experiences – ages 0–6	2.00	0.61	0.00	3.00
Positive experiences – ages 7–12	2.17	0.55	0.00	3.00
Positive experiences – ages 13–18	2.28	0.51	0.17	3.00
Abuse – Σύνολο	0.28	0.33	0.00	1.78
Abuse – ages 0–6	0.16	0.30	0.00	1.83
Abuse – ages 7–12	0.31	0.37	0.00	2.08
Abuse – ages 13–18	0.42	0.42	0.00	2.26
Traumatic life events – Total	0.28	0.24	0.00	1.58
Traumatic life events – ages 0–6	0.15	0.25	0.00	1.46
Traumatic life events – ages 7–12	0.27	0.32	0.00	1.88
Traumatic life events – ages 13–18	0.45	0.39	0.00	2.13
Family chaos – Total	0.13	0.18	0.00	1.14
Family chaos – ages 0–6	0.06	0.12	0.00	0.83
Family chaos – ages 7–12	0.09	0.16	0.00	1.06
Family chaos – ages 13–18	0.22	0.29	0.00	1.75
Distress	0.87	0.82	0.00	3.00
Inventory of Personality Organization[b]	1.89	0.44	1.06	3.52
Primitive defenses	2.22	0.60	1.00	3.94
Identity diffusion	2.28	0.64	1.00	4.57
Reality testing	1.64	0.50	1.00	3.50
Aggression	1.37	0.36	1.00	3.33
Moral values	2.00	0.63	1.00	4.88

[a] 0–3.
[b] 1–5.

4.4. Prediction of borderline personality organization from childhood trauma

Structural equation modeling for the prediction of total BPO from the four factors of childhood trauma yielded coefficients between 0.10 and 0.23, which indicate small effect sizes (see Fig. 1). More specifically, positive childhood experiences significantly and negatively predicted BPO for all age periods, whereas abuse significantly and positively predicted BPO for all age periods. Traumatic life events significantly and positively predicted BPO for ages 7–12 and 13–18. Finally, family chaos significantly and positively predicted BPO for ages 0–6, 7–12, and 13–18 Figs. 2-4.

Next, structural equation modeling was conducted to test predictions separately for each age period. With regard to the age period 0–6 (see Fig. 2), structural equation modeling for the prediction of the five factors of BPO from the four factors of childhood trauma yielded coefficients between 0.09 and 0.23, which indicate small effect sizes. More specifically, positive childhood experiences significantly and negatively predicted all factors of BPO for both genders, except primitive defenses and moral values in women. Abuse significantly and positively predicted primitive defenses and identity diffusion for men, whereas for women it predicted only moral values. Traumatic life events did not predict BPO for both genders. Family chaos significantly and positively predicted only reality testing for women.

For the age period 7–12 (see Fig. 3), structural equation modeling for the prediction of the five factors of BPO from the four factors of childhood trauma yielded coefficients between 0.08 and 0.26, which indicate small effect sizes, except for the prediction of primitive defenses by abuse, which had small to moderate effect size. More specifically, positive experiences significantly and negatively predicted, for both genders, identity diffusion, reality testing, and aggression, as well as primitive defenses for men only. Abuse significantly and positively predicted all factors of BPO for both genders, except reality testing for women. Traumatic life events significantly and positively predicted, only for men, primitive defenses, identity diffusion, reality testing, and aggression. Family chaos significantly and positively predicted, only for women, identity diffusion, reality testing, and aggression.

Finally, for the age period 13–18 (see Fig. 4), structural equation modeling for the prediction of the five factors of BPO from the four factors of childhood trauma yielded coefficients between 0.10 and 0.18, which indicate small effect sizes. More specifically, positive experiences significantly and negatively predicted primitive defenses, identity diffusion, and reality testing for both genders, as well as aggression and moral values for women only. Abuse significantly and positively predicted all factors of BPO for both genders, except reality testing for men. Traumatic life events significantly and positively predicted, only for women, primitive defenses, identity diffusion, and reality testing. Finally, family chaos significantly and positively predicted identity diffusion only for women and aggression only for men.

Table 2
Pearson r correlations between variables.

Variables	1	2	3	4	5	6	7	8
1. Positive experiences	–							
2. Abuse	−0.28***	–						
3. Traumatic life events	−0.16**	0.41***	–					
4. Family chaos	−0.24***	0.50***	0.41***	–				
5. Primitive defenses	−0.29***	0.38***	0.27***	0.25***	–			
6. Identity diffusion	−0.31***	0.32***	0.24***	0.27***	0.75***	–		
7. Reality testing	−0.32***	0.30***	0.31***	0.28***	0.63***	0.70***	–	
8. Aggression	−0.30***	0.42***	0.30***	0.37***	0.55***	0.52***	0.60***	–
9. Moral values	−0.23***	0.30***	0.21***	0.21***	0.40***	0.40***	0.44***	0.50***

** $p < .01$.
*** $p < .001$.

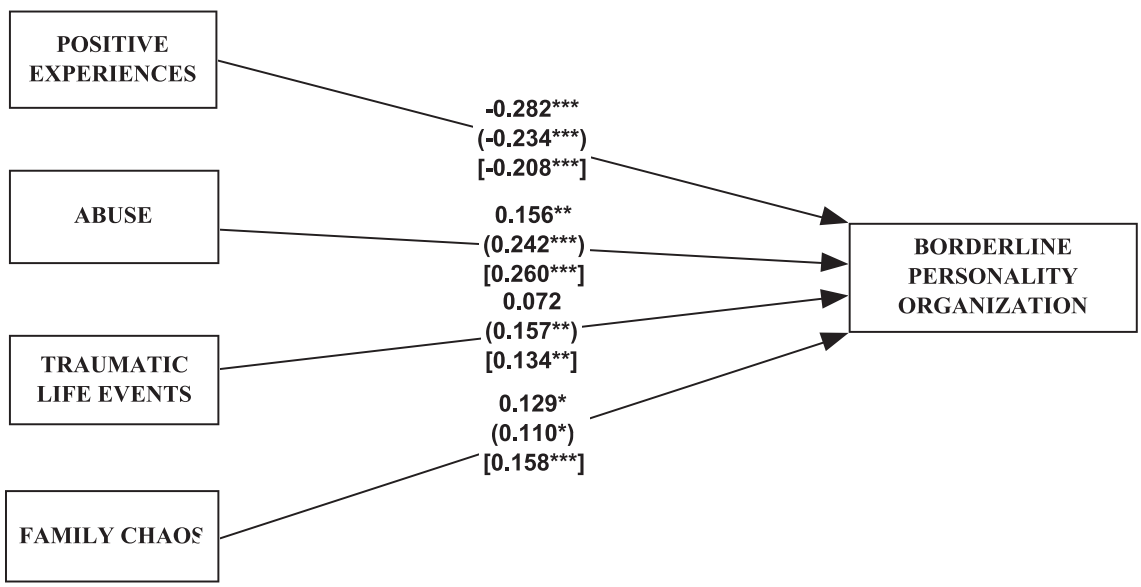

Fig. 1. Structural Equation Modeling for the Prediction of Borderline Personality Organization from Positive and Traumatic Experiences, for Each Age Period
Note. In each group of coefficients, the first one is for ages 0–6, the one in parenthesis for ages 7–12, and the one in brackets for ages 13–18.

5. Discussion

This retrospective study aimed at examining whether and to what extent various forms of childhood trauma, but also positive childhood experiences, predict BPO according to Kernberg's model, in a community sample of emerging adults (ages 18–29). Results generally supported our hypotheses. Statistically significant associations were found between childhood trauma and BPO, although these associations, as expected, were low to moderate as indicated by the rather small effect sizes. A plausible explanation is that the etiology of BPO arises from a complex interplay among genetic/neurobiological, psychodynamic, and psychosocial factors, thus, it is not determined only by childhood trauma (Kernberg, 1984, 2015; Kernberg & Michels, 2009). Moreover, the associations between childhood trauma and BPO are expected to be less strong in nonclinical or community than in clinical samples (Fuchshuber et al., 2018, 2019; Igarashi et al., 2010; see also the meta-analysis of Porter et al., 2020), which may be an additional explanation of the magnitude of predictions in this study.

5.1. Factorial structure of the measures

Confirmatory factor analyses revealed many important findings to add to the existing knowledge about the theoretical structure of the two measures included in our study. First, examination of the factorial structure of the TAQ has been largely neglected. This study showed that positive childhood experiences, that is, competence and safety, emerged as a factor distinct from traumatic experiences. Moreover, the analyses showed three constellations of traumatic experiences, in which interpersonal and non-interpersonal trauma are clearly differentiated, as it is also confirmed in recent research (Spinazzola et al., 2021). A similar, but not identical, structure was also found by Saleptsi et al. (2004) and used by Karamanolaki et al. (2016) in Greece. Of notable interest is the finding that sexual abuse was distinct from the other forms of abuse, that is, emotional and physical abuse and witnessing trauma (mainly domestic violence) and loaded on the same factor as neglect and alcohol/drug use. This finding supports the view that abuse in childhood and adolescence is likely to occur in a context of neglect and chaotic family functioning (Zanarini et al., 1989). Moreover, the fact that traumatic experiences are not distinct from one another, but rather form constellations, indicates that childhood trauma is usually a complex one (Cook et al., 2005). The relatively low internal consistency reliability of two of the four factors may be attributed to the very small number of items (i.e., only two) and the diversity of the traumatic experiences.

Another key finding is the confirmation of the five-factor theoretical structure of the IPO in a Greek community sample of university students. More specifically, primitive defenses, identity diffusion, reality testing, aggression, and moral values were identified and confirmed. The emergence of the theoretical structure of the IPO, although with the use of item parcels, is an important finding given that existing studies offer conflicting evidence on this structure (e.g., Ellison & Levy, 2012; Lenzenweger et al., 2001; Normandin et al., 2002; Preti et al., 2015; Wang et al., 2022). The total instrument and the five factors had adequate

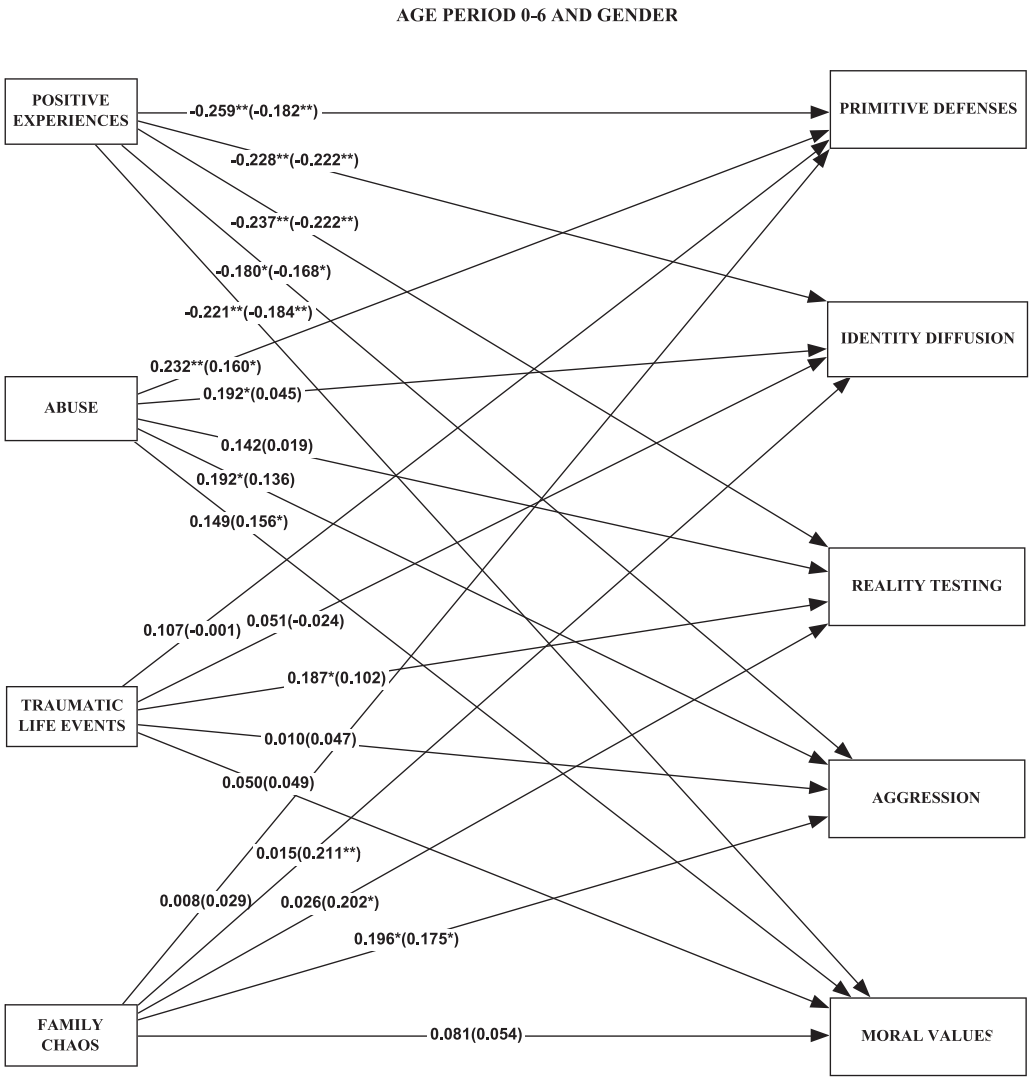

Fig. 2. Structural Equation Modeling for the Prediction of Borderline Personality Organization from Positive and Traumatic Experiences as a Function of Gender, for Ages 0–6
Note. For each variable, the first coefficient is for men and the one in parenthesis is for women.

internal consistency reliability. These findings encourage the use of the Greek version of the IPO for early screening as well as for diagnostic and research purposes in this country.

5.2. *Prediction of borderline personality organization from childhood trauma*

In general, childhood traumatic experiences had low to moderate links with BPO, as indicated by the correlations and the structural equation modeling (i.e., effect sizes). Specifically, as expected, positive experiences of childhood and adolescence, in the sense of competence and safety, were less likely in those emerging adults who exhibited higher levels of personality pathology. The protective role of competence and safety was found, in general, for all the age periods studied (i.e., 0–6, 7–12, 13–18) and for all dimensions of BPO, but especially for identity diffusion, although there were some gender differences. For example, positive experiences during adolescence seem to have a stronger protective role for women than for men regarding moral development (aggression and moral values). These results provide empirical support to the protective role of positive experiences in childhood and adolescence for BPO and especially for its essence, which is identity diffusion (Kernberg, 2006). This is a highly neglected research issue that needs further investigation, as there are very few studies examining the role of positive childhood experiences in personality disorders and especially in relation to the role played by negative experiences, and hardly any research on the role of positive childhood experiences in BPO (Gunay-Oge et al., 2020; Saleptsi et al., 2004; Skodol et al., 2007; for a review see Han et al., 2023).

Regarding traumatic experiences, a finding of significance both for Kernberg's model and for clinical work is that they were differentially associated with BPO and its various dimensions. This finding also underlines the necessity of using the full version of the IPO for screening and research purposes, especially regarding its associations with trauma. More specifically, in accordance with our hypotheses, the traumatic experience that presented the strongest links with BPO, as shown by correlations and the structural equation modeling, was abuse, which, in this study, includes both emotional and physical abuse as well as exposure to violence (mainly domestic). Abuse significantly predicted most dimensions of BPO, especially for age periods 7–12 and 13–18, compared to age period 0–6. Specifically, abuse significantly predicted primitive defenses, identity diffusion, deficits in reality testing, aggression, and superego pathology. Risk for personality pathology (i.e., primitive defenses and identity diffusion) was higher in men who reported abuse during ages 0–6 compared to women. This finding implies

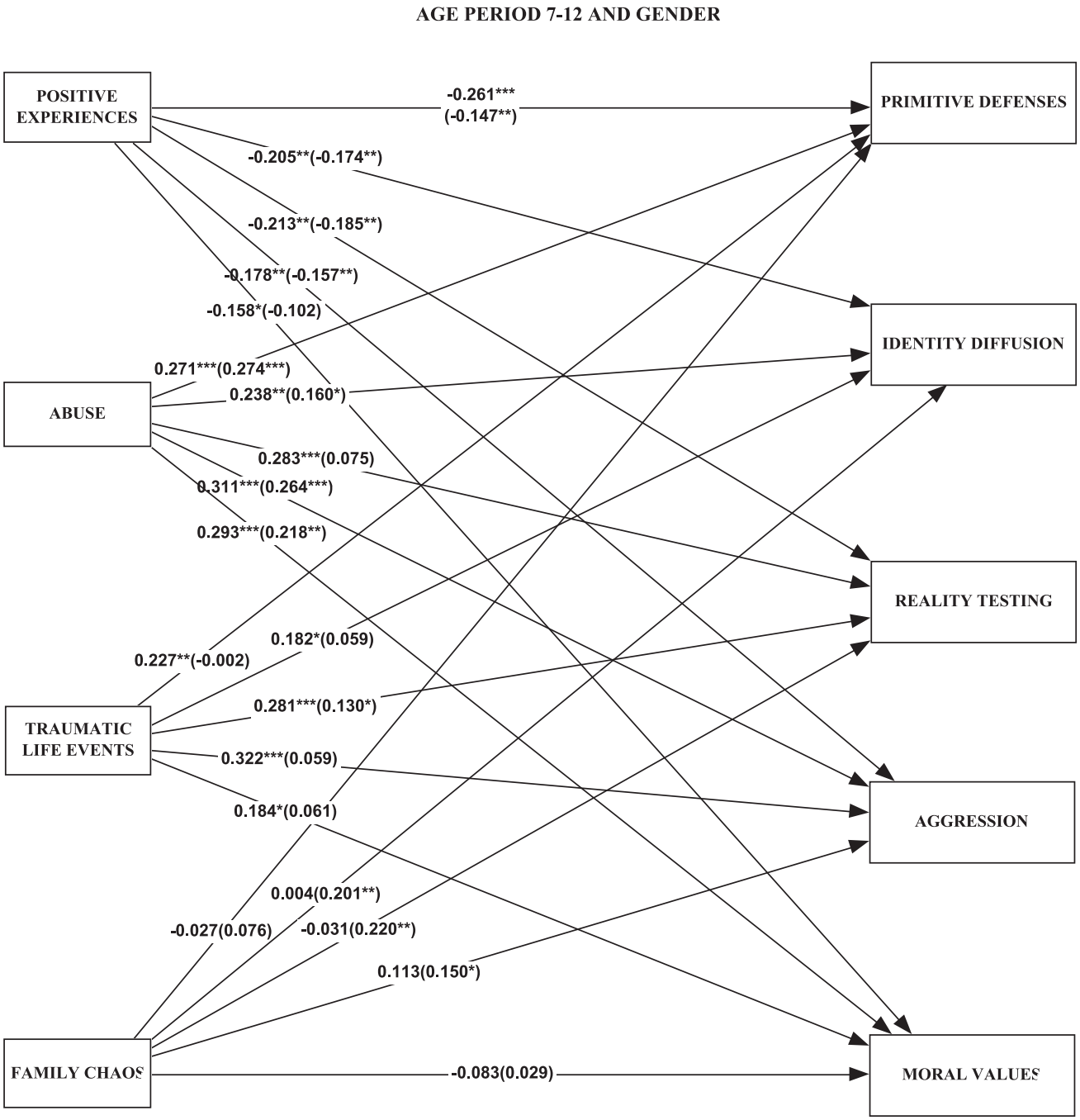

Fig. 3. Structural Equation Modeling for the Prediction of Borderline Personality Organization from Positive and Traumatic Experiences as a Function of Gender, for Ages 7–12
Note. For each variable, the first coefficient is for men and the one in parenthesis is for women.

that men may be more vulnerable in early trauma. In contrast, early trauma may be a higher risk factor for the emergence of deficits in superego formation, as reflected in the lower internalization of moral values, in women compared to men.

These findings, combined with the finding that abuse is the type of trauma mostly related to post-traumatic distress, are consistent with existing data on the associations between childhood abuse and personality pathology (see, e.g., reviews: Back et al., 2021; Bozzatello et al., 2021; Carr et al., 2013; and meta-analyses: Porter et al., 2020; Winsper et al., 2016; Yuan, Lee, Eack, et al., 2023). The findings also support Kernberg's model of internalized object relations, according to which disruptions of relationships with parental objects and especially the severe disruption caused by abuse, constitute a crucial risk factor for the formation of maladaptive self- and other-representations and of identity (Caligor & Clarkin, 2010; Clarkin et al., 2007; Kernberg, 1984).

Similarly, traumatic life events that involved separation from parents as well as other events (e.g., accident, disease, death) tended to significantly predict BPO, and more dimensions of it, when they occurred in older ages (i.e., 7–12, 13–18). These events seemed to have an especially negative effect on reality testing among both men and women. A possible explanation for this finding is that participants are less likely to retain memories of such events from the age period 0–6, especially if these events occurred to other persons and not to themselves. Similar findings, although for borderline personality disorder, have been found

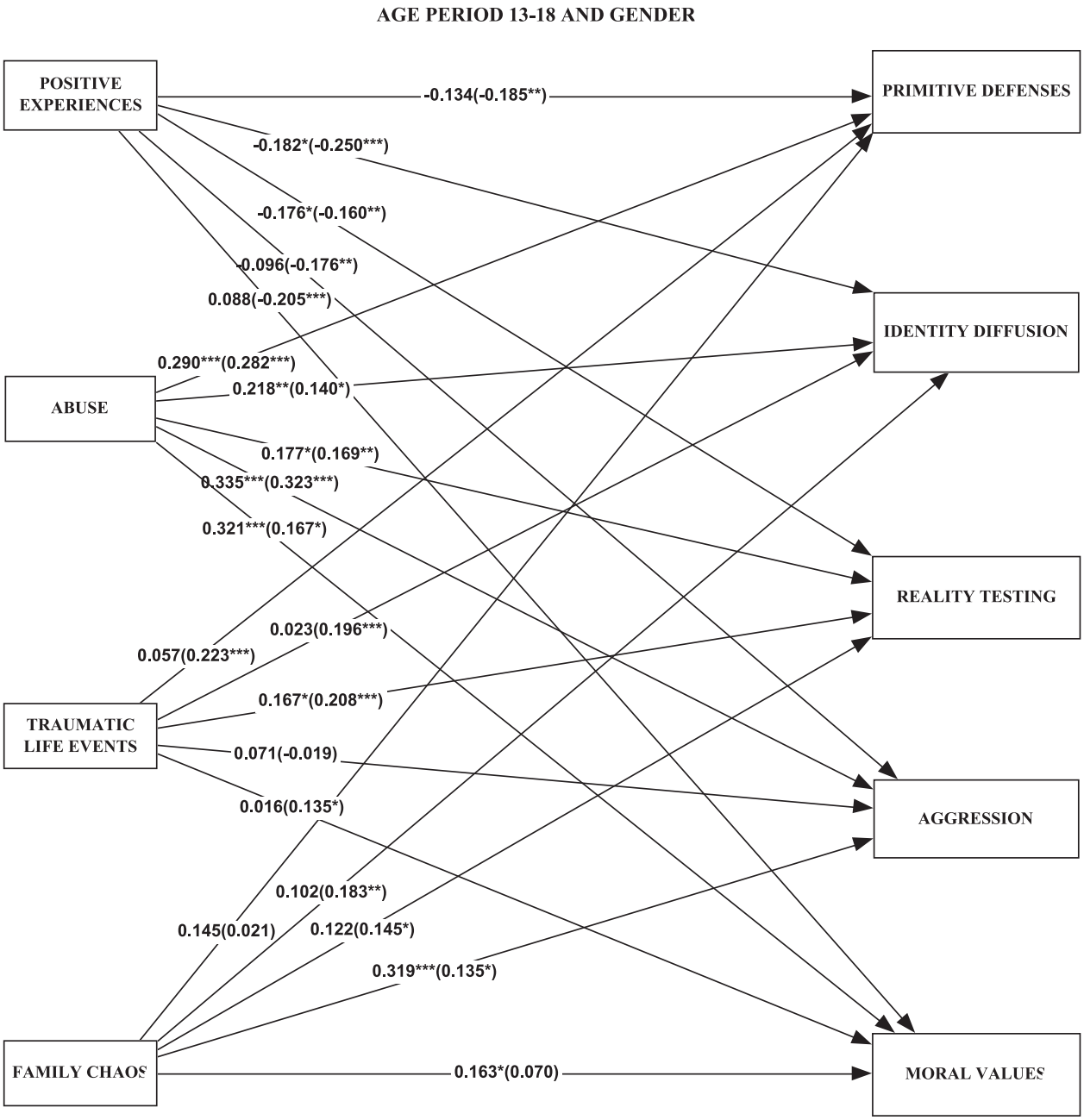

Fig. 4. Structural Equation Modeling for the Prediction of Borderline Personality Organization from Positive and Traumatic Experiences as a Function of Gender, for Ages 13–18
Note. For each variable, the first coefficient is for men and the one in parenthesis is for women.

for separation and disease during childhood (Bandelow et al., 2005; Crawford et al., 2009). It is important to note that men seem to be more negatively influenced by traumatic life events that took place in their childhood, whereas women are more influenced when these events happened during their adolescence – again this finding supports the potential higher vulnerability of men to early trauma. A possible explanation of this vulnerability stems from the differential socialization of boys and girls: from early childhood, boys are expected to exhibit greater autonomy and agency than girls when dealing with stressful situations, thus, compared to girls, they may be less likely to receive support from adults and more likely to experience harsh discipline (i.e., some type of violence) by them. However, this is only a hypothetical explanation and more investigation of gender discrepancies in this issue is needed.

Next, family dysfunction, involving sexual abuse, neglect, and alcohol and drug use, significantly predicted some dimensions of BPO, and these links were stronger for older ages. Similar findings have emerged in the literature for various personality disorders and especially for borderline personality disorder (for reviews see de Aquino Ferreira et al., 2018; Carr et al., 2013; Hutsebaut & Aleva, 2021; Martins et al., 2011; Schorr et al., 2020; Stepp et al., 2016; Yuan, Lee, Newhill, et al., 2023; and for meta-analyses: Porter et al., 2020; Winsper et al., 2016). Family dysfunction in all age periods seemed to have more adverse effects on women than men, especially regarding identity diffusion and reality testing. This can be attributed to the fact that part of the family dysfunction in this study was sexual abuse, which has been found to be more detrimental for women regarding personality pathology (de Aquino Ferreira et al., 2018; Herman et al., 1989; Paris et al., 1994;

Zanarini et al., 1997, 1989). However, we also found that men's antisocial tendencies (i.e., aggression) were aggravated when they had experienced family chaos during adolescence.

Finally, the finding that the strongest links between childhood trauma and BPO were identified when trauma had occurred in older than younger ages does not confirm our hypothesis as it is somewhat incongruent with Kernberg's model, which, due to its psychoanalytic basis, places greater emphasis on the early developmental periods. A similar finding with the use of the TAQ and its associations with personality disorders among psychiatric patients emerged in the study of Saleptsi et al. (2004). Methodological limitations, such as limited narrative memory in early ages and difficulties in recall, which may have led to under-reporting of early trauma, may explain this finding. Another explanation may be the possible reciprocal or cyclical relation between trauma and BPO. For example, difficulties in early developmental periods may lead to further difficulties in later periods, that is, to an increasing dysregulation over time, according to the cascade model (Lyons-Ruth & Brumariu, 2021). Existing research indicates that pathological personality characteristics may be present in childhood (Hecht et al., 2014) and BPO may be present in adolescence (Chanen & Thompson, 2019), rendering individuals vulnerable to various traumatic experiences (Tate et al., 2022). Thus, a one-way association from childhood trauma to BPO is not likely. Furthermore, if these early traumatic experiences subsided and were compensated by positive experiences, it is possible that their negative impact on personality organization was lessened. Indeed, in this study, positive experiences occurring during ages 0–6 predicted lower levels of all dimensions of BPO (with a few exceptions) for both men and women. This finding is consistent with Kernberg's psychodynamic model and with the aforementioned explanations. However, this is a complex issue that warrants further investigation.

5.3. Limitations and suggestions for future research

Despite its contribution, this research has certain limitations. First, childhood trauma was assessed only with the use of self-reports, therefore the results may have been affected by shared method variance. Second, this is a cross-sectional and retrospective study, thus, we cannot exclude the possibility of under-reporting (or even over-reporting) of childhood and adolescent trauma. This bias may be attributed to several factors, such as intentional concealment, difficulties in recalling early experiences, difficulties in recalling due to dissociation – a very common defensive reaction among traumatized individuals (e.g., Vonderlin et al., 2018) –, and tendency to respond negatively due to depressive affect (Baldwin et al., 2019). For example, the finding that the associations of childhood trauma with BPO were stronger when trauma occurred at older ages can be attributed, at least in part, to recall bias. However, existing research has shown that, although individuals tend to under-report their traumatic experiences, these retrospective reports are relatively reliable and do not change in the course of time (Igarashi et al., 2010). Furthermore, there are research data supporting the view that subjective perception of trauma plays a more important role in the emergence of psychopathology, compared to objective measures, such as child protection records (Baldwin & Esposti, 2021; Danese & Widom, 2020). A third limitation is that this study did not examine possible mediating variables in the relation between childhood trauma and BPO, especially given the rather small effect sizes that were found in this study. Future research could investigate such variables, for example, capacity for emotion regulation and mentalization, attachment style, or social support. Although there are many studies examining mediating variables between childhood trauma and various personality disorders, this is not the case with BPO as conceptualized by Kernberg.

In addition, in this study we did not examine each type of traumatic experience separately, but preferred to use the four factors that resulted from the confirmatory factor analysis of the TAQ. Although this analysis revealed constellations of traumatic experiences that are theoretically and clinically meaningful, the predictive value of certain traumatic experiences may have been obscured. Therefore, a more detailed examination of the impact of specific types of childhood trauma on BPO is a promising research area.

Another limitation of this study is the nonclinical and voluntary character of the sample, which is likely to account for the rather low frequency of traumatic experiences, a fact that possibly did not allow for the emergence of stronger links between the variables studied. Consequently, future research should include clinical samples as well as comparisons with nonclinical or community samples of emerging adults. However, if we consider that BPO may manifest itself with subclinical/subthreshold features in the general population of emerging adults and that it impedes adult role taking (Chanen et al., 2020), future research should further detect borderline personality pathology and examine its developmental antecedents in community samples.

Finally, the sample of this study consisted mainly of university students, who typically have a satisfactory level of functioning that is necessary for responding to academic challenges. They also tend to come from families with higher educational and socioeconomic status. Therefore, the sample cannot be considered as representative of the whole population of emerging adults.

6. Conclusion

Despite its limitations, this study contributes to the literature by highlighting the predictive value of mainly interpersonal trauma (i.e., abuse, family dysfunction) during childhood and adolescence for all dimensions of BPO, as well as the differential impact of the various traumatic experiences on BPO. These associations are low to moderate, implying the multifactorial etiology of personality pathology. It is noteworthy that these significant predictions were found in a community sample, a portion (slightly less than one-third) of which, though, had sought and received help from mental health professionals. The promotive role of childhood positive experiences was also found. The results provide empirical support for the model of internalized object relations, as conceptualized by Kernberg (1975, 1984), and underline the clinical significance of investigating emerging adults' developmental history and of working through childhood trauma. Finally, the fact that BPO places obstacles in the way to adulthood, mainly through identity diffusion and relationship impairment, necessitates future research on emerging adults' level of personality organization and its antecedents, with the aim of formulating early, developmentally-sensitive, trauma-informed, and evidence-based interventions.

CRediT authorship contribution statement

Aikaterini Malafanti: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Vasiliki Yotsidi:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization, Methodology. **Georgios Sideridis:** Writing – review & editing, Methodology, Formal analysis, Visualization. **Eleni Giannouli:** Writing – review & editing, Project administration, Methodology, Investigation. **Evangelia P. Galanaki:** Writing – review & editing, Methodology, Investigation. **Ioannis Malogiannis:** Validation, Project administration, Methodology.

Declaration of competing interest

None.

Data availability

Due to the sensitive nature of the data, they are not shared publicly, but are available upon reasonable request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.actpsy.2024.104181.

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